SOFTWARE ENGINEER TECHNICAL ASSIGNMENT

VMC Operator HMI - Startup Guidance

Build a responsive full-stack HMI for one VMC operator. After the machine is powered on, the interface must guide the operator through machine checks, tool loading, workpiece setup and final readiness before enabling a simulated operation start.

1 | Candidate assumptions and user

Define one mock scenario
Assume the quantity, operation, material, drawing revision, CNC program, fixture, work offset, required tools and ordered setup instructions. Preload these values; order creation and acceptance are not required.

Primary user	Interface principle
One VMC machine operator The operator follows the instruction currently displayed on the HMI	Show one stage at a time Use large, clear status and action controls Do not add unrelated menus or features

2 | Required operator screen sequence

Stage	What the HMI must display	Operator action
1. Machine checks	Power/control available; E-stop released; guard/door closed; no active alarm; lubrication/coolant ready; reference return complete.	Confirm each check.
2. Required tools	Tools required for the assumed operation, including tool number/type and CNC program revision.	Insert and confirm each tool.
3. Workpiece setup	Fixture, workpiece orientation, clamping instruction, material/drawing revision and work offset.	Arrange, clamp and confirm.
4. Ready review	Completed machine, tooling and workpiece checklist with a clear READY state.	Proceed to operation.
5. Operation	Operation name and simple READY, RUNNING or STOPPED status.	Start or stop the simulation.

3 | Minimal UI controls

Control	Expected simulated behavior
Confirm Check	Confirm the machine, tool or workpiece item currently displayed.
Next	Open the next stage only after every item on the current stage is confirmed.
Start Operation	READY -> RUNNING only after all arrangements are complete.
Stop Operation	RUNNING -> STOPPED and preserve the latest stage.

4 | Expected output

Expected application behavior	Submission by email
POWER ON -> MACHINE CHECKS -> TOOLS -> WORKPIECE -> READY -> RUNNING Responsive HMI, mock data, API and simple persistence Only the active instruction, progress, status and essential controls	Host the working UI at an accessible live URL Reply to the assignment email with the link Include demo login details if required Keep the link active for review